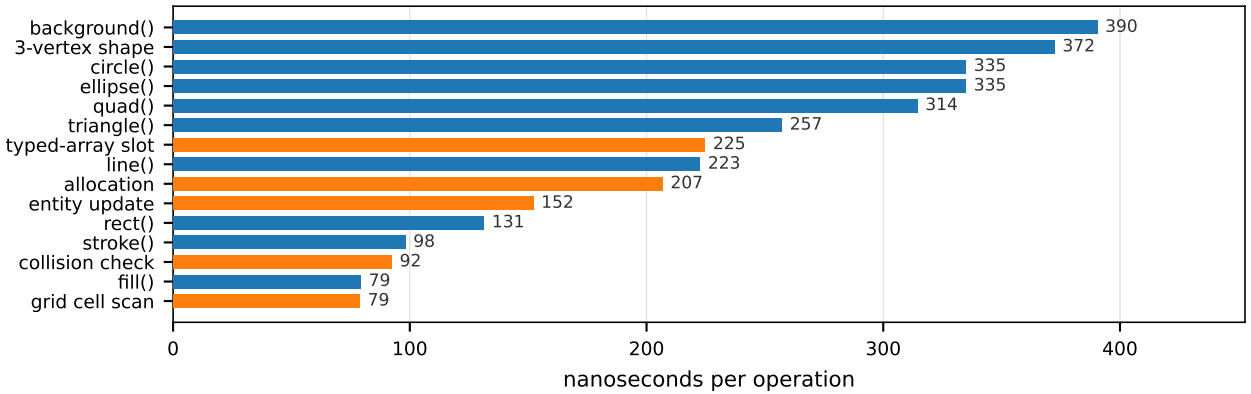
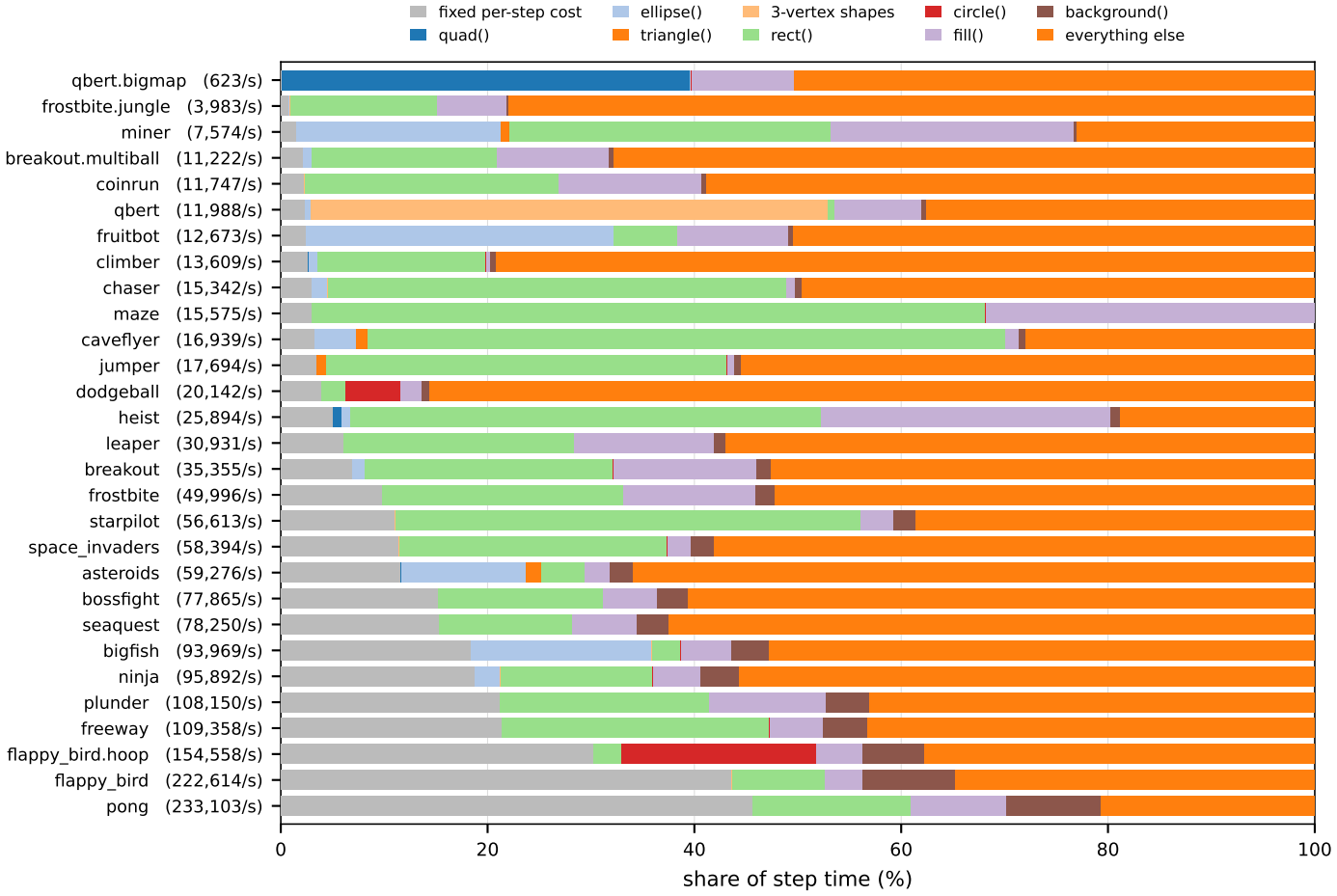

background()
3-vertex shape
circle()
ellipse()
quad()
triangle()
typed-array slot
line()
allocation
entity update
rect()
stroke()
collision check
fill()
grid cell scan
390
372
335
335
314
257
225
223
207
152
131
98
92
79
79
0
100
200
300
400
nanoseconds per operation


fixed per-step cost
quad()
ellipse()
triangle()
3-vertex shapes
rect()
circle()
fill()
background()
everything else
qbert.bigmap (623/s)
frostbite.jungle (3,983/s)
miner (7,574/s)
breakout.multiball (11,222/s)
coinrun (11,747/s)
qbert (11,988/s)
fruitbot (12,673/s)
climber (13,609/s)
chaser (15,342/s)
maze (15,575/s)
caveflyer (16,939/s)
jumper (17,694/s)
dodgeball (20,142/s)
heist (25,894/s)
leaper (30,931/s)
breakout (35,355/s)
frostbite (49,996/s)
starpilot (56,613/s)
space_invaders (58,394/s)
asteroids (59,276/s)
bossfight (77,865/s)
seaquest (78,250/s)
bigfish (93,969/s)
ninja (95,892/s)
plunder (108,150/s)
freeway (109,358/s)
flappy_bird.hoop (154,558/s)
flappy_bird (222,614/s)
pong (233,103/s)
0
20
40
60
80
100
share of step time (%)